

POLYMATHS - THE PAPERS THE LORD TAUGHT US

Polymaths - my papers and what they taught me;

Papers
+
References

This is a tale of a self taught mathematician, self taught musician, self taught ELL and linguist.

[1] Grossi, P. (2026). Principia Orthogona, Volume One: The Mathematics of Generative Transitions. Zenodo. <https://doi.org/10.5281/zenodo.19117400>

Rejected by HAL on the grounds of no affiliation / no credentials ref.
<https://hal.science/hal-05555216>

Email from HAL (Verbatim): Support by CCSD<hal-support@ccsd.cnrs.fr>

You

+1 other

Dear Author,

Thank you for your message.

We regret to inform you that your submission cannot be accepted on HAL. HAL only accepts submissions from researchers who are either enrolled in a PhD program, hold an official doctoral degree from an internationally recognized institution, or, in exceptional cases, authors with an extensive list of publications in internationally recognized peer-reviewed journals.

This policy ensures that the works deposited on HAL come from researchers affiliated with academic institutions or those with established scientific recognition (<https://about.hal.science/>).

Unfortunately, students, professionals, or independent researchers who are neither PhD candidates, doctoral degree holders, nor internationally recognized through their publications cannot be accepted, regardless of the scientific quality of their work, which is not in question.

If you are enrolled in a PhD program, hold an official doctoral degree, or have a significant bibliography in scientific journals, you may send proof of your status (enrollment certificate, diploma, ORCID profile, Google Scholar profile, etc.) so that we can re-examine your case. Otherwise, we encourage you to share your work on other open platforms, such as Zenodo, which allow all authors to make their research visible.

Thank you for your understanding.

Best regards,

The HAL Support Team

POLYMATHS - THE PAPERS THE LORD TAUGHT US

Le Ven 20 Mar 2026 12:14:29, pgrossi888@outlook.com a écrit :

> Dear HAL Support Team,

>

> Thank you for your message regarding deposit hal-05559997. I write to
> provide the clarification you requested regarding my researcher status
> and institutional affiliation, and to formally request that my deposit
> be validated under HAL's exceptional pathway for unaffiliated
> independent researchers.

>

> _____

> IDENTITY AND ORCID

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> Name: Pablo Nogueira Grossi
> Email: pablogrossi@hotmail.com
> ORCID: <https://orcid.org/0009-0000-6496-2186>
> Location: Elizabeth / Newark, New Jersey, USA

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> RESEARCHER STATUS

> _____

>

> I am an independent researcher. I do not hold a doctoral degree, and I
> am not enrolled in a doctoral program. I want to be transparent about
> this from the outset, and I believe the exceptional pathway described
> in your message – available to unaffiliated authors who can
> demonstrate recognition in the scientific community – applies to my
> case.

>

> My academic background: I studied Geology (one semester) at the
> Universidade de Brasília in 1999, and I completed a Bachelor's degree
> in Tourism in 2003. Personal circumstances – specifically, the birth
> of my son Victor Luca Jugnet Grossi on July 27, 2000 – required me to
> leave my original program of study. Victor passed away on January 4,
> 2019. The research program submitted to HAL is, in part, dedicated to
> his memory.

>

> From 2000 onward, I pursued the mathematical research that became the
> Principia Orthogona trilogy in parallel with professional life,
> including service on the staff of Banco do Brasil (one of Brazil's

POLYMATHS - THE PAPERS THE LORD TAUGHT US

- > most selective competitive placements) and, subsequently, work as
- > Academic Coordinator at UCEDA School (ESL), Elizabeth, NJ, USA — the
- > institutional address listed in the deposit.
- >
- > _____
- > THE RESEARCH AND ITS SCIENTIFIC STANDING
- > _____
- >
- > The deposit hal-05559997 is one of three companion papers constituting
- > the Principia Orthogona series, a unified mathematical framework
- > spanning:
- >
- > • Volume 1 (Topographical Orthogenesis / Generative Transitions):
- > singularity theory and Riemannian geometry, operator sequence $C \rightarrow K \rightarrow$
- > $F \rightarrow U$, Whitney A_1 – A_3 classification.
- > • Volume 2 (Generative Contact Mechanics / dm3): contact geometry,
- > operator algebra, four main theorems with explicit structural
- > stability radius.
- > • Volume 3 (Biological Instantiations): dm3 framework applied to four
- > oscillatory biological systems, three falsifiable quantitative
- > predictions.
- >
- > All three papers are typeset in formal LaTeX using the amsart document
- > class, with full theorem-proof structure, explicit MSC 2020 subject
- > classifications, and bibliographies of 40+ peer-reviewed references
- > spanning contact geometry, dynamical systems, stochastic stability,
- > and mathematical biology.
- >
- > The companion papers are also deposited on Zenodo:
- > <https://zenodo.org>
- > (searches: "Grossi Generative Contact Mechanics" and "Grossi dm3
- > operator")
- >
- > The work is conducted under the institutional umbrella of G6 LLC, a
- > research organization registered in the State of New Jersey
- > (registration date: January 3, 2025), which is currently preparing a
- > formal research infrastructure grant application to the U.S. National
- > Science Foundation (NSF Division of Mathematical Sciences),
- > referencing these preprints as prior work.
- >

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> _____
> REQUEST
> _____
>
> I respectfully request that HAL consider validating hal-05559997 under
> the exceptional pathway for unaffiliated authors, on the basis of:
>
> 1. The mathematical rigor and formal structure of the submitted work
> (amsart, full proofs, MSC classifications, 40+ citations).
> 2. Cross-platform deposition on both HAL and Zenodo, demonstrating
> commitment to open science.
> 3. ORCID profile: <https://orcid.org/0009-0000-6496-2186>
> 4. Institutional registration: G6 LLC, Newark, NJ (est. January 3,
> 2025), the organizational home of the research program.
> 5. Active NSF grant application referencing these preprints as prior
> work – confirming engagement with the formal research funding system.
> 6. Professional affiliation: Academic Coordinator, UCEDA School (ESL),
> Elizabeth, NJ – the institution listed in the deposit address field.
>
> I am in the process of requesting that UCEDA School confirm my role on
> their institutional web presence, and I am happy to provide any
> documentation they can supply.
>
> I am also happy to provide additional information, answer questions
> about the mathematical content of the work, or send any supporting
> documents to hal.support@ccsd.cnrs.fr.
>
> I understand the importance of HAL's moderation standards and I
> appreciate the care your team takes. I ask only for the opportunity to
> have this work evaluated on its scientific merits under the
> exceptional pathway your policy provides.
>
> Thank you sincerely for your time and consideration.
>
> Respectfully,
> > Pablo Nogueira Grossi
> Independent Researcher, G6 LLC | Academic Coordinator, UCEDA School
> 1180 Elizabeth Ave. Elizabeth, NJ

L'équipe support HAL

POLYMATHS - THE PAPERS THE LORD TAUGHT US

REFERENCES

- [1] Grossi, P. (2026). Principia Orthogona, Volume One: The Mathematics of Generative Transitions. Zenodo. <https://doi.org/10.5281/zenodo.19117400>
- [2] Grossi, P. (2026). Principia Orthogona — Generative Contact Mechanics - A Geometric Framework for Dissipative Systems with Structured Limit Cycles. Zenodo. <https://doi.org/10.5281/zenodo.19122168>
- [3] Grossi, P. (2026). The DM3 Operator - Explicit Toy Model and Global Dynamic Analysis. Zenodo. <https://doi.org/10.5281/zenodo.19379385>
- [4] Grossi, P. (2026). Principia Orthogona Volume Two: Contact Realization of Generative Transitions. Zenodo. <https://doi.org/10.5281/zenodo.19379473>
- [5] Grossi, P. (2026). The G6 Crystal: A dm3-Derived Architectural Form for Resonance-Stable Tall Structures. Zenodo. <https://doi.org/10.5281/zenodo.19162013>
- [6] Grossi, P. (2026). Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT). Zenodo. <https://doi.org/10.5281/zenodo.19208015>
- [7] Grossi, P. (2026). Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT): A Fruit-Fly Connectome Toy Model. Zenodo. <https://doi.org/10.5281/zenodo.19210137>
- [8] Grossi, P. (2026). The Swarm Simulator: A Dynamical Systems Model of Collective Intelligence Using the TO/TOGT Operator Pipeline. Zenodo. <https://doi.org/10.5281/zenodo.19208284>
- [9] Grossi, P. (2026). Mathematical Foundations of Multi-Orbit Identity Theory within the TO/TOGT Operator Framework. Zenodo. <https://doi.org/10.5281/zenodo.19210058>
- [10] Grossi, P. (2026). The Collatz Conjecture as a Corollary of Crystal Geometry: A Supplement to the Principia Orthogona Crystal Paper. Zenodo. <https://doi.org/10.5281/zenodo.19378742>

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Carta ao IMPA — Tres Provas : Collatz e ou Syracuse | Editor Note for Submission
260401-Grossi (Journal of Experimental Mathematics)

From: Pablo Grossi <pablogrossi@hotmail.com>

Sent: Wednesday, April 1, 2026 9:52 AM

To: editora@impa.br <editora@impa.br>

Cc: pablogrossi@hotmail.com <pablogrossi@hotmail.com>

Subject: Carta ao IMPA — Tres Provas : Collatz e ou Syracuse | Editor Note for Submission 260401-Grossi (Journal of Experimental Mathematics)

Carta ao IMPA — Três Provas : Collatz e ou Syracuse

Pablo Nogueira Grossi

Pesquisador Independente

G6 LLC · Newark, New Jersey, EUA

ORCID: 0009-0000-6496-2186

E-mail: pablogrossi@hotmail.com

1º de abril de 2026

**À Diretoria e Pesquisadores do Instituto de Matemática Pura e Aplicada –
IMPA, Rio de Janeiro – RJ - Brasil**

**Assunto: Encaminhamento de manuscrito e capítulo correspondente para
registro temporal e apreciação científica**

Prezados(as) Senhores(as),

**Tenho a honra de encaminhar ao IMPA o manuscrito “Three Proofs of the
Collatz Conjecture via Generative Topographic Contact Dynamics”,
acompanhado do capítulo correspondente do meu livro *Principia
Orthogona: The Mini-Beast*.**

**Permito-me registrar, de forma explícita, que embora esta comunicação
seja enviada em 1º de abril, data tradicionalmente associada a
brincadeiras, o conteúdo aqui apresentado é estritamente sério, técnico e
científico. A escolha da data decorre apenas do calendário editorial e não
possui qualquer conotação lúdica. A matéria tratada é de natureza
matemática rigorosa e submetida com absoluto respeito à comunidade
científica.**

Encaminho ambos os documentos pelos seguintes motivos:

1. Registro temporal (timestamp) do arcabouço teórico e dos resultados extraídos.

O manuscrito é autossuficiente e contém as demonstrações completas. Entretanto, o capítulo do livro documenta o desenvolvimento conceitual do arcabouço GTCT (Generative Transition Contact Dynamics), no qual as provas foram originalmente formuladas. Enviar ambos os materiais assegura um registro claro, transparente e datado da evolução das ideias.

2. Contextualização opcional para pesquisadores do IMPA.

O capítulo não é necessário para a verificação das demonstrações, mas pode ser útil para colegas que desejem compreender a motivação geométrica, a estrutura dinâmica e a origem dos invariantes utilizados. Sua inclusão é opcional e visa apenas oferecer contexto adicional, caso desejado.

Reitero que a avaliação científica deve recair exclusivamente sobre o manuscrito, sendo o capítulo um complemento documental para fins de clareza e arquivamento.

Envio este material ao IMPA por reconhecer o papel central que a instituição desempenha na matemática brasileira e internacional, especialmente nas áreas de sistemas dinâmicos, geometria e estruturas matemáticas profundas — campos nos quais o arcabouço GTCT se insere naturalmente.

Coloco-me à disposição para quaisquer esclarecimentos adicionais e ficarei honrado com qualquer comentário, crítica ou orientação que os pesquisadores do IMPA julgarem pertinentes.

Agradeço desde já pela atenção e pelo trabalho contínuo que o IMPA realiza em prol da ciência.

Atenciosamente,

Pablo Nogueira Grossi

Pesquisador Independente

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E-mail: pablogrossi@hotmail.com

Letter to IMPA — Submission of Manuscript + Chapter

**Pablo Nogueira Grossi G6 LLC · Newark, New Jersey, USA ORCID:
0009-0000-6496-2186 Email: pablogrossi@hotmail.com**

1 April 2026

**To the Esteemed Directors and Researchers of IMPA, Instituto de
Matemática Pura e Aplicada Rio de Janeiro, Brasil**

**Subject: Submission of Manuscript and Corresponding Book Chapter for
Timestamping and Scholarly Review**

Dear Colleagues,

**It is with great respect for IMPA's scientific tradition that I share my
manuscript *Three Proofs of the Collatz Conjecture via Generative
Topographic Contact Dynamics*, together with the corresponding chapter
from my forthcoming book *Principia Orthogona: The Mini-Beast*.**

I am sending both documents for two reasons:

**1. Scholarly timestamping of the framework and the extracted results. The
manuscript presents a self-contained mathematical treatment of the Collatz
map within the GTCT (Generative Transition Contact Dynamics) framework.
The accompanying chapter documents the broader geometric and
dynamical structure from which the proofs were originally developed.
Sharing both ensures a clear and transparent timestamp for the framework
and its mathematical consequences.**

2. Contextual clarity for IMPA's researchers. While the manuscript stands on its own, the chapter may be useful for colleagues who wish to understand the conceptual background — including the operator-chain structure, the contact-Hamiltonian formulation, and the phase-field invariants. The chapter is not required for the correctness of the manuscript; it is provided only as optional context and for archival completeness.

My intention is to place the work within IMPA's orbit of scientific dialogue, both because of the institute's central role in Brazilian mathematics and because the GTCT framework intersects with areas where IMPA has deep expertise: dynamical systems, geometry, and mathematical physics.

I would be honored if the institute considered the manuscript for informal review, discussion, or archival timestamping. Any feedback — conceptual, structural, or critical — would be deeply appreciated.

Thank you for your time, attention, and for the scientific leadership IMPA continues to provide to Brazil and to the world.

With sincere respect,

Pablo Nogueira Grossi Independent Researcher G6 LLC · Newark, New Jersey, USA ORCID: 0009-0000-6496-2186 Email: pablogrossi@hotmail.com

Editor Note for Submission 260401-Grossi

(Journal of Experimental Mathematics)

Dear Editors,

Thank you for considering my manuscript *Three Proofs of the Collatz Conjecture via Generative Topographic Contact Dynamics* (Submission ID: 260401-Grossi). Alongside the manuscript, I am submitting the corresponding chapter from my book *Principia Orthogona: The Mini-Beast* as supplementary material.

I am including this chapter for two reasons:

1. **Timestamping the broader framework.** The manuscript is fully self-contained, but the underlying geometric and dynamical framework (GTCT) was originally developed in the book chapter. Submitting both documents together provides a clear scholarly timestamp for the framework and the extracted results.
2. **Optional context for referees.** Referees who wish to understand the conceptual background, the origin of the invariants, or the motivation behind the operator-chain and contact-Hamiltonian constructions may find the chapter helpful. Its inclusion is optional and does not affect the self-contained nature of the manuscript.

Please evaluate the submission solely on the basis of the manuscript. The chapter is provided only for transparency and archival completeness.

Thank you for your time and consideration.

Sincerely, Pablo Nogueira Grossi G6 LLC · Newark, New Jersey, USA
ORCID: 0009-0000-6496-2186 Email: pablogrossi@hotmail.com

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Depois de enviar o “papel serio” para o [IMPA.br](https://impa.br), publicamos um papel mais serio no Zenodo, e seria bom mas desnecessário buscar peer-review de conjecturas. Provas apenas com a linguagem correta, o motivo pelo qual estamos desenvolvendo esta língua.

Papel (“Sério”) omitido - Provas para o Collatz dependem de conjecturas, e servem para ilustrar apenas o que poderia ser feito ao utilizar TOGT, mas isso dependerá de outrem.

Pablo Grossi

New Jersey - 4/2/2026.

- Veja “The Collatz Conjecture”, em anexo.
- Vide “The Collatz Conjecture” paper, attached.